

Homework 4

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Problem

The daily administration of the law firm faces a critical bottleneck due to its reliance on manual tools like spreadsheets for updating trials. This method consumes excessive administrative time, disorganizes documentation, and delays billing processes, which negative impacts overall efficient and case traceability. With multiple profiles integrating without a strict audit log, it becomes impossible to automatically track which user modified case file information, compromising the integrity of legal data.

The most serious deficiency of this manual management is the high risk of missing mandatory judicial deadlines. Spreadsheets lack an active alert system, preventing automatic notifications from being sent to assigned lawywer´s mobile phones when a deadline or hearing date approaches. This is compouned by operational fragmentation: users lose valuable time because they lack a global search engineindexed by the client´s name that immediately links trial data with its respective documents and evidence in PDF or Word formats.

Finally, the current infrastructure limits the mobility, agility, and security required by modern legal practice. The team lacks a multi-platform architecture that allows secure access to case files from any device (computers, tablest, or cell phones) and location (such as directly from the courthouses). Furthermore, the managed confidential informtaion lacks the necessary encryption standards, making it urgent to implement a secure system that allows for the mass migration of historical Excel data and the digitization of remaining physical archives.

Overview

The Legal Case Management System comprehensive software solution designed to transform how the firm handles its daily operations, completely leaving behind the slow and error-prone reliance on manual spreadsheets. Instead of having information scattered across different files and physical folders, this system acts as a centralized digital hub. It brings together all crucial information-client details, defendant records, judicial dependences, assigned lawyers, and real-time case statuses-into one single, easy-to-use platform.

Beyond just storing data, the system is designed to actively assist the legal team. It features a robust document management module that allows users to securely upload and preview PDF and Word files directly linked to their respective cases. One of its most powerful fearures is an automated calendar system that tracks judicial deaslines ans send proactive push notifications to the lawyers mobile devices, ensuring that critical hearing dates are never missed. Additionally, the system includes a lightning-fast global search engine indexed primarily by client name, allowing any team member to retrieve a complete case file in seconds rather than minutes.

Ultimately, the main goal of Project JC is to remove the administrative burden from the lawyers' shoulders. By automating repetitive tasks, optimizing document organization, and ensuring complete traceability through user activity logs, the software allows the legal team to focus on what truly matters: providing excellent legal counsel and winning cases

Background

In law firms, case information management is essential for maintaining proper oversight of judicial proceedings, organizing legal documentation, and providing efficient services to clients. Lawyers and legal assistants must continuously update case records, manage documents related to each legal process, and consult records associated whit clients, defendants, and judicial actions. As the number of active cases increases, these tasks become more complex and require a more complex and require a more organized approach to information management.

Currently, the law firm manages its case information using Microsoft Excel spreadsheets. Although this tool allows for the recording of basic data, it presents limitations when it comes to efficiently managing information related to clients, case files, documents, and system users. Manual updates of records consume a considerable amount of time, increase the risk of human error, and make it difficult to maintain consistent and up-to-date-information.

Another significant challenge lies in the organization of legal documentation.

Documents associated with different cases are often stored separately from the case records, making them difficult to locate and retrieve quickly. This situation ca cause delays in information searches and negatively affect the efficiency of the firms's daily operations

Additionally, it's necessary to have mechanisms that allow the identification of which user has made modification to a specific case file. The absence of a detailed change log makes it

difficult to trace actions performed within the system and limits control over stored information.

To address these issues, the development of a legal case management system is proposed to centralize information related to clients, case files, user, and associated documents. The system will facilitate the registration and retrieval of information, improve document organization, and provide a history of modifications made by authorized users, contributing to a more efficient and structured administration of legal processes.

- **Nouns**

  Priority: High
  Done: No

N°	Noun	N°	Noun
1	Activity log	26	Judicial dependence
2	Administration	27	Judicial proceeding
3	Alert system	28	Law firm
4	Audit log	29	Lawyer
5	Billing	30	Legal assistant
6	Calendar	31	Legal counsel
7	Case	32	Legal data
8	Case file	33	Location
9	Case status	34	Mobile phone
10	Change log	35	Modification
11	Client	36	Notification
12	Confidential information	37	Operations
13	Courthouse	38	Physical archive
14	Defendant	39	Platform
15	Device	40	Profile
16	Document	41	Push notification
17	Document management module	42	Record
18	Documentation	43	Search engine
19	Encryption	44	Software
20	Evidence	45	Spreadsheet
21	Excel data	46	System
22	Format	47	Team member
23	Hearing	48	Traceability
24	Infrastructure	49	Trial
25	Judicial deadline	50	User

Class Diagrams

User
-userId : int -username : String -passwordHash : String -email : String -role : Role -active : boolean
+authenticate(password : String) : boolean +changePassword(newPassword : String) : void +getDetails() : String

Role
-roleId : int -roleName : String -permissions : List<String>
+getPermissions() : List<String> +hasPermission(permission : String) : boolean

Lawyer
-lawyerId : int -name : String -professionalId : String -phone : String -specialty : String
+getAssignedCases() : List<Case> +receiveNotification(notification : Notification) : void +getDetails() : String

LegalAssistant
-assistantId : int -name : String -supervisor : Lawyer
+updateCase(legalCase : Case) : boolean +uploadDocument(document : Document) : boolean

Client
-clientId : int -name : String -idNumber : String -phone : String -email : String -address : String
+register() : boolean +update() : boolean +getDetails() : String

Defendant
-defendantId : int -name : String -idNumber : String -address : String
+register() : boolean +getDetails() : String

Case
-caseId : int -caseNumber : String -title : String -client : Client -defendant : Defendant -dependency : JudicialDependency -assignedLawyer : Lawyer -status : String -openingDate : Date -documents : List<Document> -hearings : List<Hearing>
+addDocument(document : Document) : boolean +addHearing(hearing : Hearing) : boolean +changeStatus(newStatus : String) : void +register() : boolean +getDetails() : String

JudicialDependency
-dependencyId : int -name : String -city : String -judge : String
+getDetails() : String

Document
-documentId : int -name : String -format : String -filePath : String -uploadDate : Date -uploadedBy : User
+upload() : boolean +preview() : void +download() : boolean +getDetails() : String

Hearing
-hearingId : int -legalCase : Case -date : Date -location : String -description : String
+reschedule(newDate : Date) : void +getDetails() : String

Deadline
-deadlineId : int -legalCase : Case -dueDate : Date -description : String -completed : boolean
+daysRemaining() : int +isOverdue() : boolean +markAsCompleted() : void

Calendar
-deadlines : List<Deadline> -hearings : List<Hearing>
+addDeadline(deadline : Deadline) : void +getUpcomingEvents(days : int) : List<Deadline> +checkDeadlines() : void

Notification
-notificationId : int -recipient : Lawyer -message : String -dateSent : Date -read : boolean
+send() : boolean +markAsRead() : void +getDetails() : String

AuditLog
-logId : int -user : User -legalCase : Case -action : String -dateTime : Date -details : String
+record() : void +filterByUser(user : User) : List<AuditLog> +filterByCase(legalCase : Case) : List<AuditLog>

SearchEngine
-cases : List<Case>
+searchByClient(name : String) : List<Case> +searchByCaseNumber(number : String) : Case +getCaseDocuments(legalCase : Case) : List<Document>

Invoice
-invoiceId : int -legalCase : Case -client : Client -issueDate : Date -amount : double -paid : boolean
+calculateTotal() : double +markAsPaid() : void +getDetails() : String

DataMigration
-sourceFile : String -importDate : Date -recordsImported : int
+importFromExcel(path : String) : boolean +digitizeArchive(file : String) : boolean +getSummary() : String

Encryption
-algorithm : String -key : String
+encrypt(data : String) : String +decrypt(data : String) : String